

Week 7 Report: Model Development (end) and Model Evaluation (start)

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

Student: Yashwanth

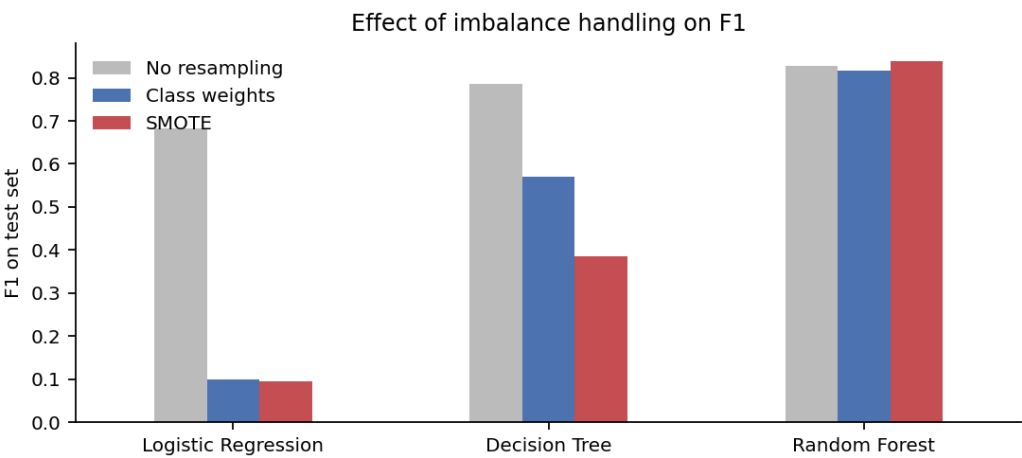
Supervisor: Dr. Prasanth Nayak

What I did this week

This week I compared three ways of handling the imbalance for every tuned model: no resampling, class weights and SMOTE. SMOTE is applied only inside the training data.

The result was not what many papers suggest. For Random Forest the strategy makes little difference (F1 between 0.818 and 0.838). For Logistic Regression, class weights and SMOTE push F1 down to about 0.10 because the model flags thousands of legitimate payments at the default threshold of 0.5. Decision Tree with SMOTE also drops sharply (F1 0.386). This agrees with Ma et al. (2026) that resampling is not always helpful.

Model	No resampling	Class weights	SMOTE
Logistic Regression	0.683	0.100	0.095
Decision Tree	0.786	0.570	0.386
Random Forest	0.829	0.818	0.838



F1 on the test set for each imbalance strategy.

Next week

Evaluate the final models in detail and test whether the differences are significant.